

Remove `wstring_convert` From C++26

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1 Abstract

The `wstring_convert` library component has been deprecated since C++17. As noted at that time, the feature is underspecified and would require more work than we wish to invest to bring it up to the desired level of quality. This paper proposes removing the deprecated convenience conversion interfaces `wstring_buffer` and `wbuffer_convert` from the C++ Standard Library.

2 Revision History

R3: March 2024 (Tokyo meeting)

- Wording updates
 - Rebased onto latest working draft, [\[N4971\]](#)
 - “will not compile” -> “may become ill-formed”
 - zombie names p3: changed “.” to “:.”

R2: September 2023 (midterm mailing)

- Removed revision history’s redundant subsection numbering
- Wording updates
 - Rebased onto latest working draft, N4958
 - Updated stable label cross-reference to C++23
 - Close all open LWG issues on the removed feature
- Removed wording concerns related to [\[P2874R2\]](#) as that paper has landed
- Retested example against MSVC with `/W3`, the IDE default
- Applied editorial recommendations
 - Cleaned up ambiguous pronouns in summary of the July 2020 SG16 review

R1: June 2023 (SG16 telecon)

- Fixed copy/paste where common text was clearly taken from another paper
- Assigned SG16 as reviewer of first resort
- Provided full library wording against current draft, N4950
- Recorded when (or if) popular library implementations first warn of deprecation
- Thanked Matt Godbolt for Compiler Explorer
- Completed SG16 review, advance to LEWG

R0: May 2023 (pre-Varna)

- Initial draft of this paper

3 Introduction

At the start of the C++23 cycle, [P2139R2] tried to review each deprecated feature of C++ to see which would benefit us to actively remove and which might now be better undeprecated. Consolidating all this analysis into one place was intended to ease the (L)EWG review process but in return gave the author so much feedback that the next revision of the paper was not completed.

For the C++26 cycle, a much shorter paper, [P2863R0], will track the overall analysis, but for features that the author wants to actively progress, a distinct paper will decouple progress from the larger paper so that the delays on a single feature do not hold up progress on all.

This paper takes up the deprecated convenience conversion interfaces `wstring_buffer` and `wbuffer_convert`.

4 History

This feature was originally proposed for C++11 by paper [N2401] and deprecated for C++17 by paper [P0618R0]. As noted at the time, the feature was underspecified and would require more work than we wished to invest to bring it up to the desired level of quality. Since then, SG16 has convened and is producing a steady stream of work to bring reliable well-specified Unicode support to C++.

Currently, four open LWG issues relate to this clause; that number would be larger, but we would prefer to see this feature removed than to keep adding issues to deprecated library features.

- [LWG2478] Unclear how `wstring_convert` uses `cvtstate`
- [LWG2479] Unclear how `wbuffer_convert` uses `cvtstate`
- [LWG2480] Error handling of `wbuffer_convert` unclear
- [LWG2481] `wstring_convert` insufficiently precise regarding “byte-error string” and so on

5 Deployment Experience

The following program, based on an example in the Standard, was tested with Godbolt Compiler Explorer to determine when (or if) libraries started warning about the deprecation.

```
#include <codecvt>
#include <iostream>
#include <locale>
#include <string>

int main() {
    std::wstring_convert<std::codecvt_utf8<wchar_t>> myconv;
    std::string mbstring = myconv.to_bytes(L"Hello\n");
    std::cout << mbstring;
}
```

- libc++: First warns in Clang 15 (2022-09-06)
- libstdc++: Does not warn in latest release
- MSVC: Warns with /W3 in MSVC 19.14, oldest available at Godbolt

5.1 Initial LEWGI review: Telecon 2020/07/13

Discussion was broadly in favor of removing from the C++23 specification and relying on library vendors to maintain source compatibility as long as needed. However, LEWGI explicitly requested to confer with SG16 in case that study group is aware of any reason to delay or to avoid removal, before proceeding with the recommendation.

5.2 SG16 review: Telecon 2020/07/22

SG16 raises concerns that the original paper that deprecated this feature ([P0618R0]) lacked a strong motivation, as that proposal was simply recording a recommendation from the LWG review when deprecating the `<codecvt>` header for [depr.locale.stdcvt]. SG16 expressed general concern that `codecvt` is not fit for its purpose, notably due to poorly specified error-handling capabilities while transcoding, and these deprecated functions do not address that underlying issue but are merely a convenience API for using that underspecified library component. While removing the `<codecvt>` header might mean there would be fewer `codecvt` facets in the C++ Standard, that deprecated API remains just as usable with user-provided `codecvt` facets as before as well as with those in the `<locale>` header. While we would like to see a replacement facility, no such proposal has been offered at this time.

Polling showed no consensus to recommend the removal for C++23 but no objection to that removal.

5.3 LEWGI consensus for C++23

SG16 has confirmed it has no objection, so the LEWGI consensus is to remove this feature from C++23.

6 Recommendation for C++26

Given vendors' propensity to provide ongoing support for these names under the Zombie Name reservations and following the LEWGI consensus for C++23, this paper proposed removing these interfaces from the C++26 Standard and closing LWG issues [LWG2478], [LWG2479], [LWG2480], and [LWG2481] as Resolved by removal of the feature per this paper.

7 C++26 Feedback

7.1 SG16 (Unicode) review

SG16 held a telecon on 07 June 2023, and reviewed this paper. The motivation given in the proposed Annex C wording was accepted, although LWG will likely want to make some updates in the wording review.

The main review comments were that one attendee observed that they had 16 uses in their code base, and all were an error that should be replaced (and will be shortly)! Another attendee performed a Github code search and found just five hits in the whole of Github.

The paper is forwarded to LEWG by unanimous consent.

7.2 LEWG initial review

The LEWG review is pending.

8 Wording

Make the following changes to the C++ Working Draft. All wording is relative to [N4971], the latest draft at the time of writing.

8.1 Add new identifiers to 16.4.5.3.2 [zombie.names].

16.4.5.3.2 [zombie.names] Zombie names

- ¹ In namespace `std`, the following names are reserved for previous standardization:

- `auto_ptr`,
- `auto_ptr_ref`,
- `binary_function`,
- ...
- `undecclare_no_pointers`,
- `undecclare_reachable`, ~~and~~
- ~~`unexpected_handler`,~~
- `wbuffer_convert`, and
- `wstring_convert`.

- ² The following names are reserved as members for previous standardization, and may not be used as a name for object-like macros in portable code:

- `argument_type`,
- `first_argument_type`,
- `io_state`,
- `open_mode`,
- `preferred`,
- `second_argument_type`,
- `seek_dir`, and
- `strict`.

- ³ The ~~`name_stoss` is reserved as a member function~~ following names are reserved as member functions for previous standardization, and may not be used as ~~a~~ names for function-like macros in portable code~~:~~

- `converted`,
- `from_bytes`,
- `stoss`, and
- `to_bytes`.

- ⁴ The header names `<ccomplex>`, `<ciso646>`, `<cstdalign>`, `<cstdbool>`, and `<ctgmath>` are reserved for previous standardization.

8.2 Update Annex C:

C.1.X Annex D: compatibility features [diff.cpp23.depr]

Change: Remove convenience interfaces `wstring_buffer` and `wbuffer_convert`.

Rationale: These features were underspecified with no clear-error reporting mechanism and were deprecated for the last three editions of this standard. Ongoing support remains at the implementers' discretion, exercising freedoms granted by 16.4.5.3.2 [zombie.names].

Effect on original feature: A valid C++ 2023 program using these interfaces may become ill-formed.

8.3 Strike all of D.23 [depr.conversions] Deprecated convenience conversion interfaces

D.23 [depr.conversions] Deprecated convenience conversion interfaces

D.23.1 [depr.conversions.general] General

- ¹ The header <locale> (30.2 [locale.syn]) has the following additions:

```
namespace std {
    template<class Codecvt, class Elem = wchar_t,
            class WideAlloc = allocator<Elem>,
            class ByteAlloc = allocator<char>>
        class wstring_convert;
    template<class Codecvt, class Elem = wchar_t,
            class Tr = char_traits<Elem>>
        class wbuffer_convert;
}
```

D.23.2 [depr.conversions.string] Class template wstring_convert

- ¹ Class template `wstring_convert` performs conversions between a wide string and a byte string. It lets you specify a code conversion facet (like class template `codecvt`) to perform the conversions, without affecting any streams or locales.

[*Example 1:* If you want to use the code conversion facet `codecvt_utf8` to output to `cout` a UTF-8 multibyte sequence corresponding to a wide string, but you don't want to alter the locale for `cout`, you can write something like:

```
wstring_convert<std::codecvt_utf8<wchar_t>> myconv;
std::string mbstring = myconv.to_bytes(L"Hello\n");
std::cout << mbstring;
```

—end example]

```
namespace std {
    template<class Codecvt, class Elem = wchar_t,
            class WideAlloc = allocator<Elem>,
            class ByteAlloc = allocator<char>>
        class wstring_convert {
        public:
            using byte_string = basic_string<char, char_traits<char>, ByteAlloc>;
            using wide_string = basic_string<Elem, char_traits<Elem>, WideAlloc>;
            using state_type = typename Codecvt::state_type;
            using int_type = typename wide_string::traits_type::int_type;

            wstring_convert() : wstring_convert(new Codecvt) {}
            explicit wstring_convert(Codecvt* pcvt);
            wstring_convert(Codecvt* pcvt, state_type state);
            explicit wstring_convert(const byte_string& byte_err,
                                    const wide_string& wide_err = wide_string());
            ~wstring_convert();

            wstring_convert(const wstring_convert&) = delete;
            wstring_convert& operator=(const wstring_convert&) = delete;
            wide_string from_bytes(char byte);
            wide_string from_bytes(const char* ptr);
            wide_string from_bytes(const byte_string& str);
```

```

        wide_string from_bytes(const char* first, const char* last);

        byte_string to_bytes(Elem wchar);
        byte_string to_bytes(const Elem* wptr);
        byte_string to_bytes(const wide_string& wstr);
        byte_string to_bytes(const Elem* first, const Elem* last);

        size_t converted() const noexcept;
        state_type state() const;

    private:
        byte_string byte_err_string;    //exposition only
        wide_string wide_err_string;    //exposition only
        Codecvt* cvtptr;                //exposition only
        state_type cvtstate;            //exposition only
        size_t cvtcount;                //exposition only
    };
}

```

- ² The class template describes an object that controls conversions between wide string objects of class `basic_string<Elem, char_traits<Elem>, WideAlloc>` and byte string objects of class `basic_string<char, char_traits<char>, Alloc>`. The class template defines the types `wide_string` and `byte_string` as synonyms for these two types. Conversion between a sequence of `Elem` values (stored in a `wide_string` object) and multibyte sequences (stored in a `byte_string` object) is performed by an object of class `Codecvt`, which meets the requirements of the standard code-conversion facet `codecvt<Elem, char, mbstate_t>`.

- ³ An object of this class template stores:

- `byte_err_string` — a byte string to display on errors
- `wide_err_string` — a wide string to display on errors
- `cvtptr` — a pointer to the allocated conversion object (which is freed when the `wstring_convert` object is destroyed)
- `cvtstate` — a conversion state object
- `cvtcount` — a conversion count

```
size_t converted() const noexcept;
```

- ⁴ *Returns:* `cvtcount`.

```

wide_string from_bytes(char byte);
wide_string from_bytes(const char* ptr);
wide_string from_bytes(const byte_string& str);
wide_string from_bytes(const char* first, const char* last);

```

- ⁵ *Effects:* The first member function converts the single-element sequence `byte` to a wide string. The second member function converts the null-terminated sequence beginning at `ptr` to a wide string. The third member function converts the sequence stored in `str` to a wide string. The fourth member function converts the sequence defined by the range `[first, last)` to a wide string.

- ⁶ In all cases:

- If the `cvtstate` object was not constructed with an explicit value, it is set to its default value (the initial conversion state) before the conversion begins. Otherwise it is left unchanged.
- The number of input elements successfully converted is stored in `cvtcount`.

- ⁷ *Returns:* If no conversion error occurs, the member function returns the converted wide string. Otherwise, if the object was constructed with a wide-error string, the member function returns the wide-error string. Otherwise,

the member function throws an object of class `range_error`.

```
state_type state() const;
```

⁸ *Returns:* `cvtstate`.

```
byte_string to_bytes(Elem wchar);
byte_string to_bytes(const Elem* wptr);
byte_string to_bytes(const wide_string& wstr);
byte_string to_bytes(const Elem* first, const Elem* last);
```

⁹ *Effects:* The first member function converts the single-element sequence `wchar` to a byte string. The second member function converts the null-terminated sequence beginning at `wptr` to a byte string. The third member function converts the sequence stored in `wstr` to a byte string. The fourth member function converts the sequence defined by the range `[first, last)` to a byte string.

¹⁰ In all cases:

- If the `cvtstate` object was not constructed with an explicit value, it is set to its default value (the initial conversion state) before the conversion begins. Otherwise it is left unchanged.
- The number of input elements successfully converted is stored in `cvtcount`.

¹¹ *Returns:* If no conversion error occurs, the member function returns the converted byte string. Otherwise, if the object was constructed with a byte-error string, the member function returns the byte-error string. Otherwise, the member function throws an object of class `range_error`.

```
explicit wstring_convert(Codecvt* pcvt);
wstring_convert(Codecvt* pcvt, state_type state);
explicit wstring_convert(const byte_string& byte_err,
    const wide_string& wide_err = wide_string());
```

¹² *Preconditions:* For the first and second constructors, `pcvt` is not null.

¹³ *Effects:* The first constructor stores `pcvt` in `cvtptr` and default values in `cvtstate`, `byte_err_string`, and `wide_err_string`. The second constructor stores `pcvt` in `cvtptr`, `state` in `cvtstate`, and default values in `byte_err_string` and `wide_err_string`; moreover the stored state is retained between calls to `from_bytes` and `to_bytes`. The third constructor stores new `Codecvt` in `cvtptr`, `state_type()` in `cvtstate`, `byte_err` in `byte_err_string`, and `wide_err` in `wide_err_string`.

```
~wstring_convert();
```

¹⁴ *Effects:* `delete cvtptr`.

D.23.3 [depr.conversions.buffer] Class template `wbuffer_convert`

¹ Class template `wbuffer_convert` looks like a wide stream buffer, but performs all its I/O through an underlying byte stream buffer that you specify when you construct it. Like class template `wstring_convert`, it lets you specify a code conversion facet to perform the conversions, without affecting any streams or locales.

```
namespace std {
    template<class Codecvt, class Elem = wchar_t, class Tr = char_traits<Elem>>
        class wbuffer_convert : public basic_streambuf<Elem, Tr> {
        public:
            using state_type = typename Codecvt::state_type;
            wbuffer_convert() : wbuffer_convert(nullptr) {}
            explicit wbuffer_convert(streambuf* bytebuf,
                                    Codecvt* pcvt = new Codecvt,
                                    state_type state = state_type());
```

```

~wbuffer_convert();

wbuffer_convert(const wbuffer_convert&) = delete;
wbuffer_convert& operator=(const wbuffer_convert&) = delete;

streambuf* rdbuf() const;
streambuf* rdbuf(streambuf* bytebuf);

state_type state() const;
private:
    streambuf* bufptr;          //exposition only
    Codecvt* cvtptr;           //exposition only
    state_type cvtstate;       //exposition only
};
}

```

² The class template describes a stream buffer that controls the transmission of elements of type `Elem`, whose character traits are described by the class `Tr`, to and from a byte stream buffer of type `streambuf`. Conversion between a sequence of `Elem` values and multibyte sequences is performed by an object of class `Codecvt`, which shall meet the requirements of the standard code-conversion facet `codecvt<Elem, char, mbstate_t>`.

³ An object of this class template stores:

- `bufptr` — a pointer to its underlying byte stream buffer
- `cvtptr` — a pointer to the allocated conversion object (which is freed when the `wbuffer_convert` object is destroyed)
- `cvtstate` — a conversion state object

```
state_type state() const;
```

⁴ *Returns:* `cvtstate`.

```
streambuf* rdbuf() const;
```

⁵ *Returns:* `bufptr`.

```
streambuf* rdbuf(streambuf* bytebuf);
```

⁶ *Effects:* Stores `bytebuf` in `bufptr`.

⁷ *Returns:* The previous value of `bufptr`.

```
explicit wbuffer_convert(
    streambuf* bytebuf,
    Codecvt* pcvt = new Codecvt,
    state_type state = state_type());
```

⁸ *Preconditions:* `pcvt` is not null.

⁹ *Effects:* The constructor constructs a stream buffer object, initializes `bufptr` to `bytebuf`, initializes `cvtptr` to `pcvt`, and initializes `cvtstate` to `state`.

```
~wbuffer_convert();
```

¹⁰ *Effects:* `delete cvtptr`.

8.4 Update cross-reference for stable labels for C++23

Cross-references from ISO C++ 2023

All clause and subclause labels from ISO C++ 2023 (ISO/IEC 14882:2023, *Programming Languages — C++*) are present in this document, with the exceptions described below.

container.gen.reqmts *see*

 container.requirements.general

depr.arith.conv.enum *removed*

depr.codecvrt.syn *removed*

[depr.conversions](#) *removed*

[depr.conversions.buffer](#) *removed*

[depr.conversions.general](#) *removed*

[depr.conversions.string](#) *removed*

depr.default allocator *removed*

depr.locale.stdcvt *removed*

depr.locale.stdcvt.general *removed*

depr.locale.stdcvt.req *removed*

depr.res.on.required *removed*

depr.string.capacity *removed*

mismatch *see* [alg.mismatch](#)

8.5 Resolve open library issues

The following library issues should be resolved as NAD as they no longer apply to the C++ Standard due to the removal of the feature.

- [\[LWG2478\]](#) Unclear how `wstring_convert` uses `cvtstate`
- [\[LWG2479\]](#) Unclear how `wbuffer_convert` uses `cvtstate`
- [\[LWG2480\]](#) Error handling of `wbuffer_convert` unclear
- [\[LWG2481\]](#) `wstring_convert` insufficiently precise regarding “byte-error string” and so on

9 Acknowledgements

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Thanks again to Matt Godbolt for maintaining Compiler Explorer, the best public resource for C++ compiler and library archaeology, especially when researching the history of deprecation warnings!

Thanks to Lori Hughes for reviewing this paper and providing editorial feedback.

10 References

- [LWG2478] Jonathan Wakely. Unclear how `wstring_convert` uses `cvtstate`.
<https://wg21.link/lwg2478>
- [LWG2479] Jonathan Wakely. Unclear how `wbuffer_convert` uses `cvtstate`.
<https://wg21.link/lwg2479>
- [LWG2480] Jonathan Wakely. Error handling of `wbuffer_convert` unclear.
<https://wg21.link/lwg2480>
- [LWG2481] Jonathan Wakely. `wstring_convert` should be more precise regarding “byte-error string” etc.
<https://wg21.link/lwg2481>
- [N2401] P.J. Plauger. 2007-09-03. Code Conversion Facets for the Standard C++ Library.
<https://wg21.link/n2401>
- [N4971] Thomas Köppe. 2023-12-18. Working Draft, Programming Languages — C++.
<https://wg21.link/n4971>
- [P0618R0] Alisdair Meredith. 2017-03-02. Deprecating `<codecvt>`.
<https://wg21.link/p0618r0>
- [P2139R2] Alisdair Meredith. 2020-07-15. Reviewing Deprecated Facilities of C++20 for C++23.
<https://wg21.link/p2139r2>
- [P2863R0] Alisdair Meredith. 2023-05-19. Review Annex D for C++26.
<https://wg21.link/p2863r0>
- [P2874R2] Alisdair Meredith. 2023-06-12. Mandating Annex D.
<https://wg21.link/p2874r2>